MTH 137(B) Dr. James Ashe
Exam 2 will be be given on March 13, 2013

2.1 Functions

- Determine if a given relation is a function.
  - From a graph; **HW 3-4.**
  - From a list of coordinates; **HW 5-6, 9-12**
  - From an equation; **HW 7-8**
- Determine the domain and range of a relation.
  - From a graph. *see the quiz*
  - From a list; **HW 9-10**
- Understanding and working with function notation, i.e., $f(x)$.
  - From a list; **HW 11-12**
  - From an equation; **HW 13-14**
- Find the average rate of change; **HW 15-16**
- Find the difference quotient; **HW 17-19**

2.2: Graphing functions

- Understand how to read and write interval notation, e.g., $1 \leq x < 5$ means x is in $[1, 5)$.
- Sketch a relation from an equation, then give its domain and range.
  - Common functions like lines, quadratics, and absolute values; **HW 1-3, 5**
  - Piecewise functions; **HW 6-8, 15-17**
- Identify when a function is decreasing, increasing, or constant for all type of functions including piecewise functions; **HW 10-14**

2.3: Function transformations.

- Understand and graph functions using the parent function and basic transformations. **HW 1-20**
  - Parent functions include: $f(x) = x$, $f(x) = x^2$, $f(x) = \sqrt{x}$, and $f(x) = |x|$.
  - Transformations:
    - $f(x \pm h)$: " + " $\Rightarrow$ *Left* and " − " $\Rightarrow$ *Right*
    - $f(x) \pm k$: " + " $\Rightarrow$ *Up* and " − " $\Rightarrow$ *Down*
    - $a \cdot f(x)$: $a > 0 \Rightarrow$ *normal* and $a < 0 \Rightarrow$ "*flip upside down*"
    - $a \cdot f(x)$: $|a| > 1 \Rightarrow$ *Narrow* and $|a| < 1 \Rightarrow$ *Wide*

2.4 Function operations.

- Add, subtract, multiply, and divide functions; **HW 1-8**
  - This includes finding the general function and the value at specific points, e.g., $f - g$ and $(f - g)(2)$.
  - You may also need to find the domain of the new function.
- Function compositions; $(f \circ g)(x) = f(g(x))$; **HW 8-10, 12-13, 16-19**

2.5 Inverse function

- Determine if a function is $1-1$/invertible, from a list or graph; **HW 1-9**
- Find the inverse function f^{-1}; **HW 10, 12, 16-20**
- Sketch an inverse function from a graph, or determine if two graphs are inverses of each other; **HW 13-16**
- Understand inverse function notation.